CHAPTER 2

DATA REPRESENTATION AND BOOLEAN ALGEBRA

CHAPTER ESSENCE AT A GLANCE:
Master the mathematical foundation of digital computers encompassing positional number systems, two's complement integer arithmetic, Boolean logic laws, and universal logic gate circuits.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Radix and Positional Weighting:** Any number is represented as a polynomial sum of positional digits multiplied by ascending powers of the number system base or radix.
- **Two's Complement Binary Representation:** Negative numbers are encoded by inverting all bits and adding one, enabling unified binary adder circuits to perform both addition and subtraction.
- **Boolean Duality and Simplification:** Digital logic functions operate on two-state binary variables governed by axioms and theorems like De Morgan's laws to minimize logic circuit complexity.
- **Universal Gate Circuit Synthesis:** NAND and NOR gates can synthesize all basic Boolean operations (AND, OR, NOT), enabling entire digital processing architectures to be constructed from a single gate type.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\overline{A+B}=\overline{A}\cdot\overline{B} \quad | \quad \overline{A\cdot B}=\overline{A}+\overline{B}$$

De Morgan's First and Second Laws

$$\text{Two's Complement}(N)=\text{One's Complement}(N)+1$$

Two's Complement Negative Representation

$$A\oplus B=\overline{A}B+A\overline{B} \quad (\text{XOR Sum Bit})$$

Exclusive OR Logic Gate Function

$$N_{\text{Decimal}}=\sum_{i=-m}^{n-1} d_i\times r^i \quad (r=\text{Radix})$$

Positional Radix Expansion Formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** In 2's complement subtraction, discard the final end-around carry generated from the most significant bit (MSB).
- **Alert:** In De Morgan's theorem proofs, always include truth table columns showing intermediate complements (not just inputs and final expressions).
- **Alert:** Hexadecimal conversions must accurately substitute decimal values 10 through 15 with letters A through F.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State and prove De Morgan's First Law using a truth table. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Statement:** The complement of a logical sum is equal to the logical product of the individual complements: (A + B)' = A' . B'. **[1 Mark]**

• **Truth Table Verification:** For inputs (A,B) = (0,0), (0,1), (1,0), (1,1), evaluate A+B gives (0,1,1,1) so (A+B)' gives (1,0,0,0). Evaluating A' gives (1,1,0,0) and B' gives (1,0,1,0), so A'.B' gives (1,0,0,0). Columns are identical. **[1 Mark]**

2. Why are NAND and NOR gates called Universal Gates? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Universal Gate Definition:** A universal gate is a logic gate that can implement any Boolean function or basic gate (AND, OR, NOT) without requiring any other gate type. **[1 Mark]**

• **Realization:** By connecting inputs together, NAND acts as NOT; by inverting NAND output, we get AND; using De Morgan's law, NAND with inverted inputs gives OR. Hence NAND and NOR are universal. **[1 Mark]**

3. Convert the decimal number 53.625 into its equivalent Binary and Hexadecimal representations. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Integer Part (53):** Divide 53 repeatedly by 2: $\frac{53}{2}$ = 26 (rem 1), $\frac{26}{2}$ = 13 (rem 0), $\frac{13}{2}$ = 6 (rem 1), $\frac{6}{2}$ = 3 (rem 0), $\frac{3}{2}$ = 1 (rem 1), $\frac{1}{2}$ = 0 (rem 1). Reading upward gives 110101 in binary. **[1 Mark]**

• **Fractional Part (0.625):** Multiply repeatedly by 2: 0.625 × 2 = 1.250 (int 1), 0.250 × 2 = 0.500 (int 0), 0.500 × 2 = 1.000 (int 1). Reading downward gives .101. Total Binary = 110101.101. **[1 Mark]**

• **Hexadecimal Conversion:** Group into 4 bits: 0011 0101 . 1010 = 35.A in Hexadecimal. Boxed Result: 110101.101 in binary and 35.A in hexadecimal. **[1 Mark]**

4. Perform the subtraction of 28 from 45 using 8-bit Two's Complement method. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Represent +45 and +28 in 8-bit Binary:** +45 = 00101101. +28 = 00011100. **[1 Mark]**

• **Find -28 in 2's Complement:** 1's complement of 00011100 is 11100011. Adding 1 gives 2's complement: 11100100. **[1 Mark]**

• **Binary Addition:** 00101101 (+45) + 11100100 (-28) = 1 00010001. Discard the end carry: result is 00010001 (Decimal +17). Boxed Result: 00010001 (+17). **[1 Mark]**

5. Explain character encoding standards: ASCII, ISCII, and Unicode with their bit lengths and applications. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **ASCII (American Standard Code for Information Interchange):** 7-bit standard encoding 128 characters (alphabets, digits, symbols), with extended 8-bit version encoding 256 characters. **[1 Mark]**

• **ISCII (Indian Standard Code for Information Interchange):** 8-bit encoding standard designed specifically for Indian languages including Malayalam, Hindi, Tamil, retaining ASCII compatibility. **[1 Mark]**

• **Unicode:** Universal encoding standard (UTF-8, UTF-16, UTF-32) capable of representing characters from every written language and mathematical symbol globally. **[1 Mark]**

6. Simplify the Boolean expression F = AB'C + A'B'C + ABC + AB'C' using Boolean algebraic laws. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Grouping Terms):** Group terms with common factors: F = B'C(A + A') + A(BC + B'C'). **[1 Mark]**

• **Step 2 (Applying Inverse Law):** Since A + A' = 1, the first term simplifies to B'C(1) = B'C. **[1 Mark]**

• **Step 3 (Rearranging & Factoring):** Write F = AB'C + AB'C' + A'B'C + ABC = AB'(C + C') + BC(A + A') = AB'(1) + BC(1) = AB' + BC. **[1 Mark]**

• **Step 4 (Final Boxed Form):** The simplified minimal expression is F = AB' + BC (or B'C + AB). **[1 Mark]**

7. Draw the logic circuit diagram for a Half Adder and write its Boolean equations and truth table. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Half Adder Concept:** Combinational circuit that adds two single-bit binary inputs (A, B) producing Sum and Carry outputs. **[1 Mark]**

• **Boolean Equations:** Sum = A XOR B = A'B + AB' ; Carry = A AND B = AB. **[1 Mark]**

• **Circuit Components:** Implemented using one 2-input XOR gate for Sum and one 2-input AND gate for Carry. Boxed Result: Sum = A XOR B, Carry = A.B. **[1 Mark]**